Clarifications on the Reproducibility of Table 1 Data (Published Article)

If you are trying to exactly reproduce the lattice parameter values described in **Table 1** of the corresponding scientific article using the scripts in this repository, please pay attention to the operational and methodological details below:

1. Origin of the Differences Found

The minor variations between the results generated by some versions of the code and the data published in the article stem from **updates to the peak filtering routine** and the **use of datasets (hkl reflections) refined manually** during the algorithm's development phase.

2. Specific Cases and Datasets Used

- **Strontium Carbonate ($SrCO_3$) - Orthorhombic:**

The original raw file extracted from the ICSD contains 128 reflections. After eliminating redundant peaks caused by selection rules and multiplicity, this volume was reduced.

The standard deviation values reported in Table 1 correspond strictly to the dataset containing **66 reflections**. The script in the first version of the code used a subset of 33 reflections (which yielded higher local precision, but lower sampling). The most recent version of the code has been updated to process and import the corrected files directly via CSV.

- **Strontium Silicate ($SrSiO_3$) - Monoclinic:**

For low-symmetry systems (such as monoclinic), computational time is directly influenced by the chosen Miller indices. Indices with a higher number of zeros drastically reduce matrix processing time.

The execution that generated the exact approximate results in Table 1 used the following manual configuration of 7 peaks (reflections and 2θ angles):

picos = [15.46, 17.53, 24.93, 27.47, 33.03, 38.62, 43.87]

h = [2, 0, 0, 2, 2, 1, 2]

k = [0, 0, 2, 0, 0, 3, 2]

l = [0, 2, 0, 2, -4, 0, 3]

- **Zinc Oxide (ZnO) - Hexagonal:**

In the first testing version of the code, only 7 reflections were manually entered (which made the process more prone to typing errors). The final validation file that backed the publication was expanded and consolidated with **10 reflections**, using the following set:

picos = [31.78, 34.44, 36.27, 47.56, 56.62, 59.58, 62.89, 66.41, 67.98, 69.12]

h = [1, 0, 0, 0, 2, 1, 0, 2, 2, 2]

k = [-1, 0, 1, 1, -1, -2, 1, -2, -1, -2]

l = [0, 2, -1, -2, 0, 1, -3, 0, -2, 1]

3. Code Evolution and Next Steps

The main reason for the initial mismatch was the transition from an old code (where peak vectors were replaced manually inside the script itself, as in the case of the initial test with LiCl for the cubic system and after with Y_2O_3 for the cubic system) to an **automated modular architecture based on CSV file importing**.

The current versions of the software now allow the user to predefine the selection criterion (such as limiting the calculation to the most intense peaks for time optimization in complex structures). The dispersion data σ should be interpreted as an **internal geometric consistency index** of the mathematical arrangement, and not as absolute uncertainty of experimental measurement.

In addition, I provide the same CSV files of all tested compounds, with the old code and the current code, for a larger 2θ range (conditions: 5-120°, step 0.01°, λ = 1.5418 Å). Thus, the user defines how many reflections and which reflections they will test.

The change we implemented, following Mac's guidance, was applied exclusively to the monoclinic ($SrSiO_3$) and triclinic [$Mg_3(PO_4)_2$] systems. For these two systems, we removed SymPy entirely in favor of using Mac's direct linear matrix function (solve_subset).

For the orthorhombic system ($SrCO_3$), your original code continues to use the SymPy solver (sp.solve); however, it is now optimized and significantly faster, thanks to the preemptive algebraic cancellation routine [.cancel()] that we previously added to your script. This allows SymPy to perform the calculations without causing the loop to freeze.

I am immensely grateful to Lloyd Herndon McDonald (https://independent.academia.edu/HerndonMac) for the email discussions that helped clarify aspects of the data used as a basis for testing our model.